

Labor Protokoll

Abgabedatum: 19.10.2025

Redundantes Routing mit OSPF

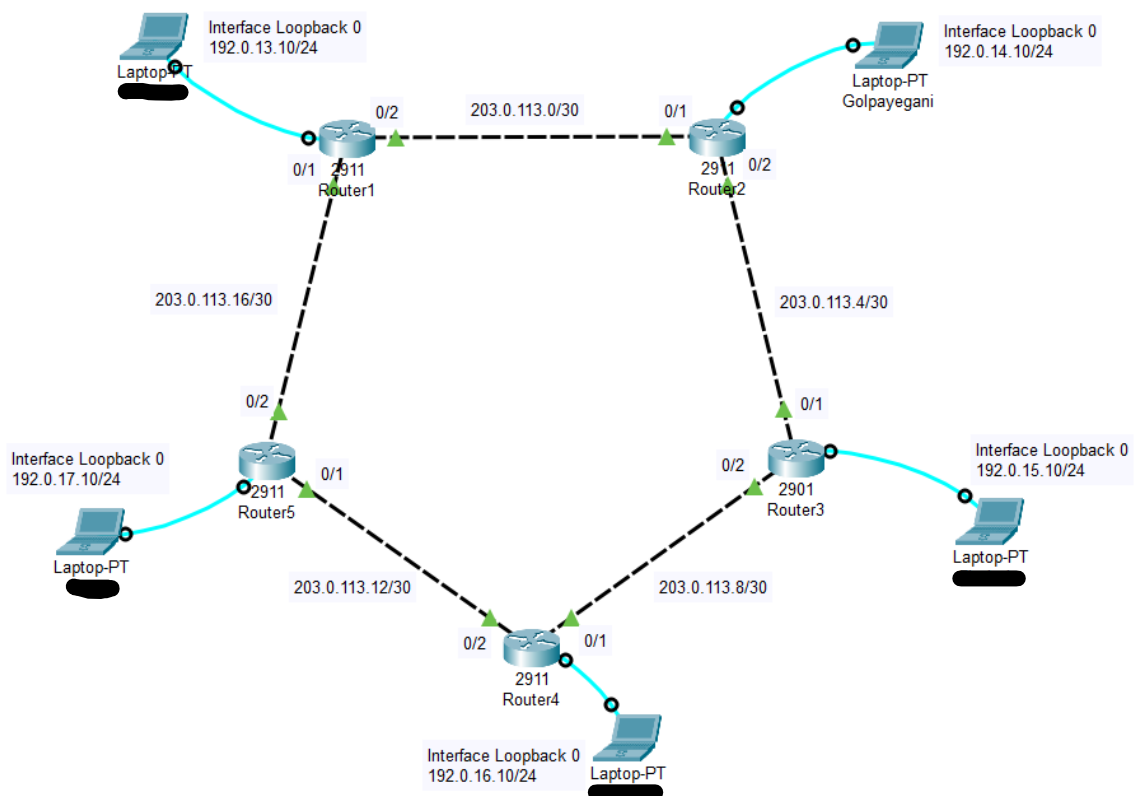
Gruppe:

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[REDACTED] (17)

Ausgangslage

In der gegebenen Netzwerktopologie sind mehrere Router in einer vermaschten Struktur miteinander verbunden. Als Routingprotokoll wird OSPF verwendet, um die automatische Verteilung und Erreichbarkeit der Routen zwischen allen Routern sicherzustellen. Ziel ist eine vollständige Konnektivität zwischen allen Geräten und Loopback-Netzen.

Topologie:



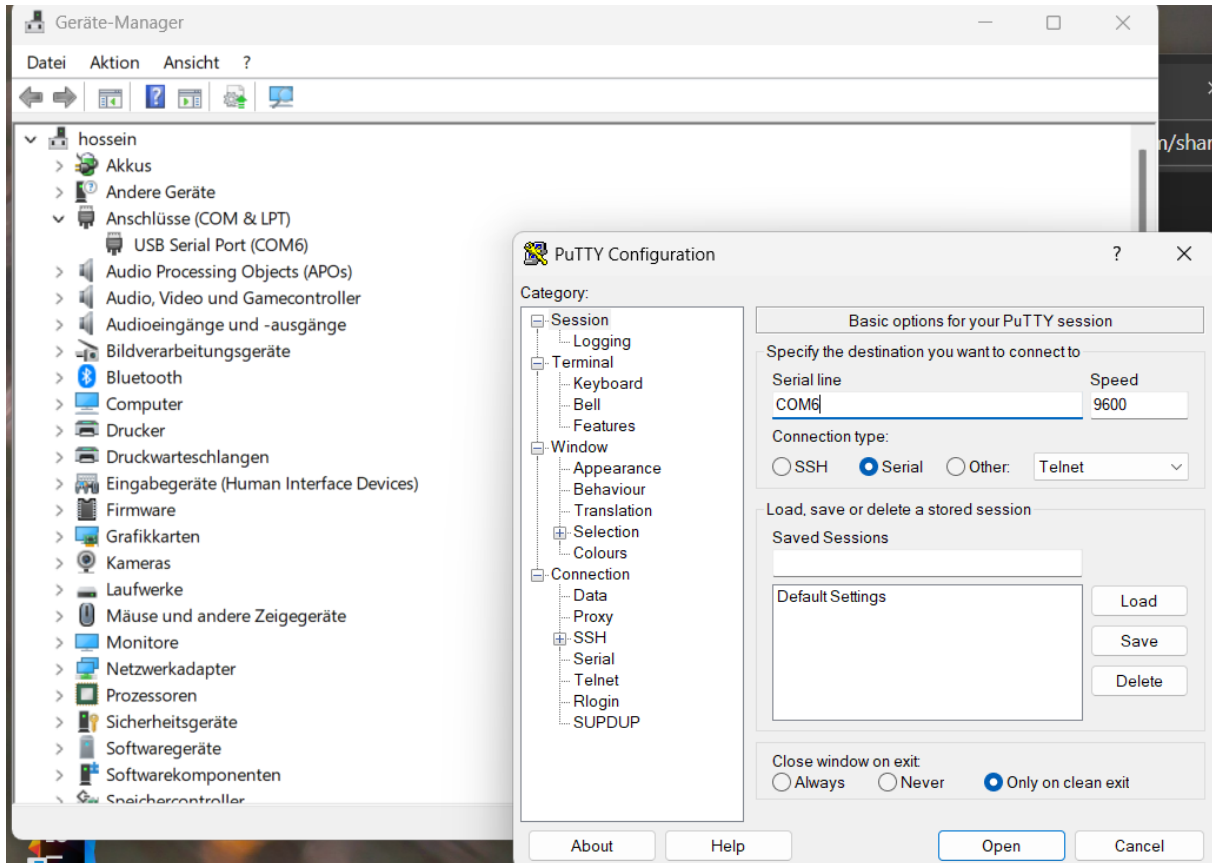
Adresstabelle:

Verwenden Sie Ihre KN für die LAN-Netzwerke folgendermaßen:

Gerät	Interface	IP Adresse	Subnet
R1	Loop 0	192.0.13.10	255.255.255.0
	Fa 0/1	203.0.113.18	255.255.255.252
	Fa 0/2	203.0.113.1	255.255.255.252
R2	Loop 0	192.0.14.10	255.255.255.0
	Fa 0/1	203.0.113.2	255.255.255.252
	Fa 0/2	203.0.113.5	255.255.255.252
R3	Loop 0	192.0.15.10	255.255.255.0
	Fa 0/1	203.0.113.6	255.255.255.252
	Fa 0/2	203.0.113.9	255.255.255.252
R4	Loop 0	192.0.16.10	255.255.255.0
	Fa 0/1	203.0.113.10	255.255.255.252
	Fa 0/2	203.0.113.13	255.255.255.252
R5	Loop 0	192.0.17.10	255.255.255.0
	Fa 0/1	203.0.113.14	255.255.255.252
	Fa 0/2	203.0.113.17	255.255.255.252

1. Verbindung mit dem Router über die serielle Schnittstelle

Über den Geräte-Manager wurde der USB Serial Port (COM6) identifiziert. In PuTTY wurde als Verbindungstyp Serial, die Schnittstelle COM6 eingestellt. Damit wurde die Verbindung zur Konsole des Routers hergestellt.



2. Grundkonfiguration des Routers

Zuerst wurde der Routername geändert und die Domain Lookup Funktion deaktiviert, um unnötige DNS-Anfragen bei Tippfehlern zu vermeiden.

Anschließend wurden die IP-Adressen für die Interfaces GigabitEthernet0/1, GigabitEthernet0/2 und Loopback0 vergeben und die Interfaces mit no shutdown aktiviert.

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```
COM6 - PuTTY
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#interface GigabitEthernet0/1
R2(config-if)#ip address 203.0.113.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)#
Oct 16 16:19:27.639: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
Oct 16 16:19:30.267: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
Oct 16 16:19:31.267: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R2(config-if)#exit
R2(config)#interface GigabitEthernet0/2
R2(config-if)#ip address 203.0.113.5 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)#exit
Oct 16 16:20:47.883: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down
R2(config)#interface Loopback0
R2(config-if)#
Oct 16 16:21:30.363: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R2(config-if)# ip address 192.0.14.10 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)#exit
```

```
Compiled Tue 20-Mar-12 18:57 by prod_rel_team
Oct 16 18:05:22.239: %SNMP-5-COLDSTART: SNMP agent on host Router is undergoing a cold start
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#no ip dom
R1(config)#no ip domain-loo
R1(config)#no ip domain-lookup
R1(config)#int gi0/1
R1(config-if)#des
R1(config-if)#description Link to R5 gi0/2
R1(config-if)#ip address 203.0.113.18 255.255.255.252
R1(config-if)#no sh
R1(config-if)#exit
R1(config)#
Oct 16 18:07:00.803: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
R1(config)#
Oct 16 18:07:04.243: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to up
Oct 16 18:07:05.243: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R1(config)#int gi0/2
R1(config-if)#des
R1(config-if)#description Link to R2 gi0/1
R1(config-if)#ip address 203.0.113.1 255.255.255.252
R1(config-if)#no sh
R1(config-if)#exit
Oct 16 18:08:20.735: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to
R1(config-if)#exit
R1(config)#int loo
R1(config)#int loopback0
R1(config-if)#ip addr
Oct 16 18:08:49.755: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed stat
R1(config-if)#ip address 192.0.13.10 255.255.255
Oct 16 18:09:09.251: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
Oct 16 18:09:10.251: %LINEPROTO-5-UPDOWN: Line protocol on Interface Gigab
R1(config-if)#ip address 192.0.13.10 255.255.255.0
R1(config-if)#no sh
R1(config-if)#exit
```

```

% Connection number required
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R4
R4(config)#no ip do
R4(config)#no ip domain-look
R4(config)#no ip domain-lookup
R4(config)#int gi0/1
R4(config-if)#desc
R4(config-if)#description Link_to_R3_gi0/2
R4(config-if)#ip address 203.0.113.10 255.255.255.252
R4(config-if)#no sh
R4(config-if)#no shutdown
R4(config-if)#
Oct 16 17:56:21.951: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
R4(config-if)#exit
R4(config)#int gi0/2
R4(config-if)#des
R4(config-if)#description Link_to_R5_gi0/1
R4(config-if)#ip address 203.0.113.13 255.255.255.252
R4(config-if)#no sh
R4(config-if)#
Oct 16 17:58:11.063: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to downex
Oct 16 17:58:14.815: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up
Oct 16 17:58:15.815: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, chan
ged state to
R4(config-if)#exit
R4(config)#int loo
R4(config)#int loopback0
R4(config-if)#ip
Oct 16 17:58:29.259: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state
to
R4(config-if)#ip address 192.0.16.10 255.255.255.0

```

```

R5(config)#interface gi 0/1
R5(config-if)#des
R5(config-if)#description lin
R5(config-if)#description Link
R5(config-if)#description Link_to_R4_gi0/2
R5(config-if)#ip add
R5(config-if)#ip address 203.0.113.14 255.255.255.252
R5(config-if)#no sh
R5(config-if)#no shutdown

```

```

R5(config)#interface gi0/2
R5(config-if)#des
R5(config-if)#description Linke_to_R4_gi0/2
R5(config-if)#ip add
R5(config-if)#desc
R5(config-if)#description Linke_to_R1_gi0/2
R5(config-if)#ip ad
R5(config-if)#ip address 203.0.113.17 255.255.255.252
R5(config-if)#no shut
R5(config-if)#no shutdown
R5(config-if)#exit

```

3. OSPF-Konfiguration

Nach der Grundkonfiguration wurde OSPF mit Prozess-ID 1 aktiviert. Als Router-ID wurde die IP-Adresse der Loopback-Schnittstelle verwendet.

Über die network-Befehle wurden die direkt angeschlossenen Netze in OSPF eingebunden und der Area 0 zugeordnet.

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```
R2(config)#router ospf 1
R2(config-router)#router-id 192.0.14.10
R2(config-router)# network 203.0.113.0 0.0.0.255 area 0
R2(config-router)#network 192.0.14.0 0.0.0.255 area 0
Oct 16 16:23:01.287: %OSPF-5-ADJCHG: Process 1, Nbr 192.0.13.10 on GigabitEthernet0/1 from LO
ADING to FULL, Loading Done
R2(config-router)#network 192.0.14.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#end
R2#
Oct 16 16:24:00.243: %SYS-5-CONFIG_I: Configured from console by console
R2#wr
Building configuration...
[OK]
```

```
R1(config)#router ospf 1
R1(config-router)#router-id 192.0.13.10
R1(config-router)#net
R1(config-router)#network 203.0.113.0 0.0.0.255 ar
R1(config-router)#network 203.0.113.0 0.0.0.255 area 0
R1(config-router)#net
R1(config-router)#network 192.0
Oct 16 18:11:01.415: %OSPF-5-ADJCHG: Process 1, Nbr 192.0.17.10 on GigabitEthernet0/1 from LO
ADING to FULL,
R1(config-router)#ne
```

```
R3(config-if)#ip address 192.0.15.10 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#
R3(config-if)#router ospf 1
R3(config-router)#router-id 192.0.15.10
R3(config-router)#network 203.0.113.0 0.0.0.255 area 0
R3(config-router)#network 192.0.
Oct 16 18:37:25.759: %OSPF-5-ADJCHG: Process 1, Nbr 192.0.16.10 on GigabitEthern
et0/2 from LOADING to FULL, Loading Done1
% Incomplete command.

R3(config-router)#network 192.0.15.0 0.0.0.255 area 0
R3(config-router)#no shutdown
R3(config-router)#
R3(config-router)#exit
R3(config)#end
R3#
Oct 16 18:38:28.963: %SYS-5-CONFIG_I: Configured from console by consolewr
Building configuration...
[OK]
R3#ping 192.0.17.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.17.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R3#
```

```

R5(config)#router os
R5(config)#router ospf 1
R5(config-router)#rout
R5(config-router)#router-id 192.0.17.10
R5(config-router)#net
R5(config-router)#network 203.0.113.0.0.0.255 are
R5(config-router)#network 203.0.113.0 0.0.0.255 area 0
R5(config-router)#netw
R5(config-router)#network 192.0.17.0 0.0.0.255 are
R5(config-router)#network 192.0.17.0 0.0.0.255 area 0

```

4. Überprüfung der Interfaces

Mit show ip interface brief wurde kontrolliert, ob die Interfaces korrekt konfiguriert und aktiv sind.

GigabitEthernet0/1 und 0/2 sowie Loopback0 standen nach der Konfiguration auf Status up/up, was auf eine erfolgreiche Aktivierung hinweist.

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```

R2#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0       unassigned      YES unset  administratively down down
GigabitEthernet0/1       203.0.113.2     YES manual up           up
GigabitEthernet0/2       203.0.113.5     YES manual up           up
Serial0/0/0              unassigned      YES unset  administratively down down
Serial0/0/1              unassigned      YES unset  administratively down down
Loopback0                192.0.14.10     YES manual up           up
R2#

```

```

Oct 16 19:08:57.863: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
R3#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down
GigabitEthernet0/0       unassigned      YES unset  administratively down down
GigabitEthernet0/1       203.0.113.6     YES manual up           up
GigabitEthernet0/2       203.0.113.9     YES manual up           up
Serial0/0/0              unassigned      YES unset  administratively down down
Serial0/0/1              unassigned      YES unset  administratively down down
Loopback0                192.0.15.10     YES manual up           up
R3#show ip ospf neighbor

```

```
R4#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
Embedded-Service-Engine0/0 unassigned      YES unset    administratively down down
GigabitEthernet0/0       unassigned      YES unset    administratively down down
GigabitEthernet0/1       203.0.113.10    YES manual   up          up
GigabitEthernet0/2       203.0.113.13    YES manual   up          up
Serial0/0/0              unassigned      YES unset    administratively down down
Serial0/0/1              unassigned      YES unset    administratively down down
Loopback0                192.0.16.10     YES manual   up          up
R4#
```

```
R5#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Embedded-Service-Engine0/0 unassigned      YES unset    administratively down down
GigabitEthernet0/0       unassigned      YES unset    administratively down down
GigabitEthernet0/1       203.0.113.14    YES manual   up          up
GigabitEthernet0/2       203.0.113.17    YES manual   up          up
Serial0/0/0              unassigned      YES unset    administratively down down
Serial0/0/1              unassigned      YES unset    administratively down down
Loopback0                192.0.17.10     YES manual   up          up
R5#ping 192.0.15.10
```

5. OSPF-Nachbarschaften prüfen

Der Befehl `show ip ospf neighbor` zeigt, dass der Router erfolgreich OSPF-Nachbarschaften zu anderen Routern aufgebaut hat.

Dabei werden die Nachbarn, die Schnittstellen sowie der aktuelle OSPF-Zustand angezeigt.

Der Zustand FULL bedeutet, dass die Adjazenz vollständig aufgebaut wurde. Zustände wie 2WAY zeigen, dass zwar eine Nachbarschaft besteht, aber keine vollständige Adjazenz (z. B. bei DROTHER-Routern).

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```
R2#show ip ospf neighbor
Neighbor ID    Pri   State           Dead Time   Address        Interface
192.0.15.10    1     2WAY/DROTHER     00:00:39   203.0.113.6    GigabitEthernet0/2
192.0.13.10    1     FULL/DR          00:00:33   203.0.113.1    GigabitEthernet0/1
R2#
Oct 16 16:32:49.243: %OSPF-5-ADJCHG: Process 1, Nbr 192.0.15.10 on GigabitEthernet0/2 from LOADING to FULL, Loading Done
```

```
R3#show ip ospf neighbor
Neighbor ID    Pri   State           Dead Time   Address        Interface
192.0.16.10    1     FULL/DR          00:00:34   203.0.113.10    GigabitEthernet0/2
192.0.14.10    1     2WAY/DROTHER     00:00:36   203.0.113.5     GigabitEthernet0/1
R3#
Oct 16 19:09:37.863: %OSPF-5-ADJCHG: Process 1, Nbr 192.0.14.10 on GigabitEthernet0/1 from LOADING to FULL,
```

```

Neighbor ID      Pri   State           Dead Time   Address      Interface
192.0.17.10      1    FULL/DR         00:00:37    203.0.113.14 GigabitEthernet0/2
192.0.15.10      1    FULL/BDR         00:00:38    203.0.113.9  GigabitEthernet0/1
R4#

```

6. OSPF-Datenbank anzeigen

Mit `show ip ospf database` wurde überprüft, welche Router und Netze im OSPF-Bereich 0 bekannt sind.

In der Ausgabe sind die Link States aller fünf Router sowie der gemeinsam genutzten Netze enthalten. Das bestätigt, dass die Routen korrekt im OSPF ausgetauscht wurden.

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```

R2#
R2#show ip ospf database

        OSPF Router with ID (192.0.14.10) (Process ID 1)

        Router Link States (Area 0)

Link ID        ADV Router    Age      Seq#           Checksum Link count
192.0.13.10    192.0.13.10   798      0x80000004    0x0085E7 3
192.0.14.10    192.0.14.10   209      0x80000005    0x00CCB3 3
192.0.15.10    192.0.15.10   139      0x80000006    0x00C5A5 3
192.0.16.10    192.0.16.10   258      0x80000005    0x00A0B8 3
192.0.17.10    192.0.17.10   898      0x80000010    0x004DEE 3

        Net Link States (Area 0)

Link ID        ADV Router    Age      Seq#           Checksum
203.0.113.1    192.0.13.10   798      0x80000001    0x0096DF
203.0.113.6    192.0.15.10   210      0x80000001    0x006805
203.0.113.10   192.0.16.10   258      0x80000002    0x004D18
203.0.113.14   192.0.17.10   391      0x80000002    0x00342A
203.0.113.17   192.0.17.10   898      0x80000001    0x00F06E
R2#

```

R1>show ip ospf database

OSPF Router with ID (192.0.13.10) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
192.0.13.10	192.0.13.10	794	0x80000004	0x0085E7	3
192.0.14.10	192.0.14.10	207	0x80000005	0x00CCB3	3
192.0.15.10	192.0.15.10	136	0x80000006	0x00C5A5	3
192.0.16.10	192.0.16.10	254	0x80000005	0x00A0B8	3
192.0.17.10	192.0.17.10	893	0x80000010	0x004DEE	3

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
203.0.113.1	192.0.13.10	794	0x80000001	0x0096DF
203.0.113.6	192.0.15.10	209	0x80000001	0x006805
203.0.113.10	192.0.16.10	254	0x80000002	0x004D18
203.0.113.14	192.0.17.10	387	0x80000002	0x00342A
203.0.113.17	192.0.17.10	893	0x80000001	0x00F06E

R1#

Link ID	ADV Router	Age	Seq#	Checksum	Link count
192.0.13.10	192.0.13.10	809	0x80000004	0x0085E7	3
192.0.14.10	192.0.14.10	221	0x80000005	0x00CCB3	3
192.0.15.10	192.0.15.10	149	0x80000006	0x00C5A5	3
192.0.16.10	192.0.16.10	265	0x80000005	0x00A0B8	3
192.0.17.10	192.0.17.10	907	0x80000010	0x004DEE	3

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
203.0.113.1	192.0.13.10	809	0x80000001	0x0096DF
203.0.113.6	192.0.15.10	220	0x80000001	0x006805
203.0.113.10	192.0.16.10	265	0x80000002	0x004D18
203.0.113.14	192.0.17.10	400	0x80000002	0x00342A
203.0.113.17	192.0.17.10	907	0x80000001	0x00F06E

R4#

7. Erreichbarkeit testen (Ping)

Pings zu allen Loopback-Adressen der anderen Router waren erfolgreich (100 % Erfolgsrate). Auch der erweiterte Ping mit der

eigenen Loopback-Adresse als Quelle verlief erfolgreich.

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```
R2#
R2#ping 192.0.16.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.16.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R2#ping 192.0.17.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.17.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R2#ping 192.0.13.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.13.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R2#ping 192.0.15.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.15.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
```

```
R2#ping
Protocol [ip]: 192.0.15.10
% Unknown protocol - "192.0.15.10 ", type "ping ?" for help
R2#ping
Protocol [ip]:
Target IP address: 192.0.15.10
Repeat count [5]:
Datagram size [100]:
Timeout in seconds [2]:
Extended commands [n]: y
Source address or interface: 192.0.14.10
Type of service [0]:
Set DF bit in IP header? [no]:
Validate reply data? [no]:
Data pattern [0xABCD]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Sweep range of sizes [n]:
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.15.10, timeout is 2 seconds:
Packet sent with a source address of 192.0.14.10
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R2#
```

```

[OK]
R1#
R1#ping 192.0.17.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.17.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#ping 192.0.16.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.16.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#ping 192.0.15.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.15.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
R1#ping 192.0.14.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.14.10, timeout is 2 seconds:
.....
Oct 16 18:12:40.135: %OSPF-5-ADJCHG: Process 1, Nbr 192.0.14.10 on GigabitEthernet0/2 from LO
ADING to FULL, Loading Done.
Success rate is 0 percent (0/5)
R1#ping 192.0.14.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.0.14.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1#

```

8. Routingtabelle überprüfen

Mit `show ip route ospf` wurde die OSPF-Routingtabelle angezeigt. Alle fremden Loopback-Netze wurden korrekt als OSPF-Routen (O) eingetragen und über die passenden Next Hops und Interfaces erreichbar gemacht.

```

R2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    192.0.13.0/32 is subnetted, 1 subnets
O       192.0.13.10 [110/2] via 203.0.113.1, 00:15:48, GigabitEthernet0/1
    192.0.15.0/32 is subnetted, 1 subnets
O       192.0.15.10 [110/2] via 203.0.113.6, 00:06:01, GigabitEthernet0/2
    192.0.16.0/32 is subnetted, 1 subnets
O       192.0.16.10 [110/3] via 203.0.113.6, 00:06:01, GigabitEthernet0/2
    192.0.17.0/32 is subnetted, 1 subnets
O       192.0.17.10 [110/3] via 203.0.113.1, 00:15:48, GigabitEthernet0/1
    203.0.113.0/24 is variably subnetted, 7 subnets, 2 masks
O       203.0.113.8/30 [110/2] via 203.0.113.6, 00:06:01, GigabitEthernet0/2
O       203.0.113.12/30 [110/3] via 203.0.113.6, 00:06:01, GigabitEthernet0/2
                                     [110/3] via 203.0.113.1, 00:15:48, GigabitEthernet0/1
--More--

```

9. Traceroute testen

Mit traceroute wurde der Weg zu verschiedenen Zielen überprüft.

Die Ausgaben zeigen die erwarteten Pfade über die jeweiligen Next Hops, was die korrekte Funktion des Routings bestätigt.

```
R2#
R2#traceroute 192.0.15.10
Type escape sequence to abort.
Tracing the route to 192.0.15.10
VRF info: (vrf in name/id, vrf out name/id)
  1 203.0.113.6 0 msec * 0 msec
R2#traceroute 192.0.13.10
Type escape sequence to abort.
Tracing the route to 192.0.13.10
VRF info: (vrf in name/id, vrf out name/id)
  1 203.0.113.1 0 msec * 0 msec
R2#traceroute 192.0.16.10
Type escape sequence to abort.
Tracing the route to 192.0.16.10
VRF info: (vrf in name/id, vrf out name/id)
  1 203.0.113.6 0 msec 0 msec 0 msec
  2 203.0.113.10 0 msec * 0 msec
R2#traceroute 192.0.17.10
Type escape sequence to abort.
Tracing the route to 192.0.17.10
VRF info: (vrf in name/id, vrf out name/id)
  1 203.0.113.1 0 msec 0 msec 0 msec
  2 203.0.113.17 0 msec * 0 msec
R2#
```

Kosten

Die Kosten zum Loopback des übernächsten Routers ergeben sich aus der Summe der OSPF-Linkkosten entlang des Pfads. Jeder Gigabit-Link hat standardmäßig einen Kostenwert von 1, da die OSPF-Referenzbandbreite bei Cisco 100 Mbit/s beträgt. Zusätzlich zählt die Loopback-Schnittstelle selbst ebenfalls 1.

In unserem Fall ergeben sich 3 Kostenpunkte (zwei Verbindungen + eine Loopback), was mit dem Eintrag in der Routingtabelle übereinstimmt.

```
R2#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    192.0.13.0/32 is subnetted, 1 subnets
O      192.0.13.10 [110/2] via 203.0.113.1, 00:15:48, GigabitEthernet0/1
    192.0.15.0/32 is subnetted, 1 subnets
O      192.0.15.10 [110/2] via 203.0.113.6, 00:06:01, GigabitEthernet0/2
    192.0.16.0/32 is subnetted, 1 subnets
O      192.0.16.10 [110/3] via 203.0.113.6, 00:06:01, GigabitEthernet0/2
    192.0.17.0/32 is subnetted, 1 subnets
O      192.0.17.10 [110/3] via 203.0.113.1, 00:15:48, GigabitEthernet0/1
    203.0.113.0/24 is variably subnetted, 7 subnets, 2 masks
O      203.0.113.8/30 [110/2] via 203.0.113.6, 00:06:01, GigabitEthernet0/2
O      203.0.113.12/30 [110/3] via 203.0.113.6, 00:06:01, GigabitEthernet0/2
                   [110/3] via 203.0.113.1, 00:15:48, GigabitEthernet0/1
--More--
```